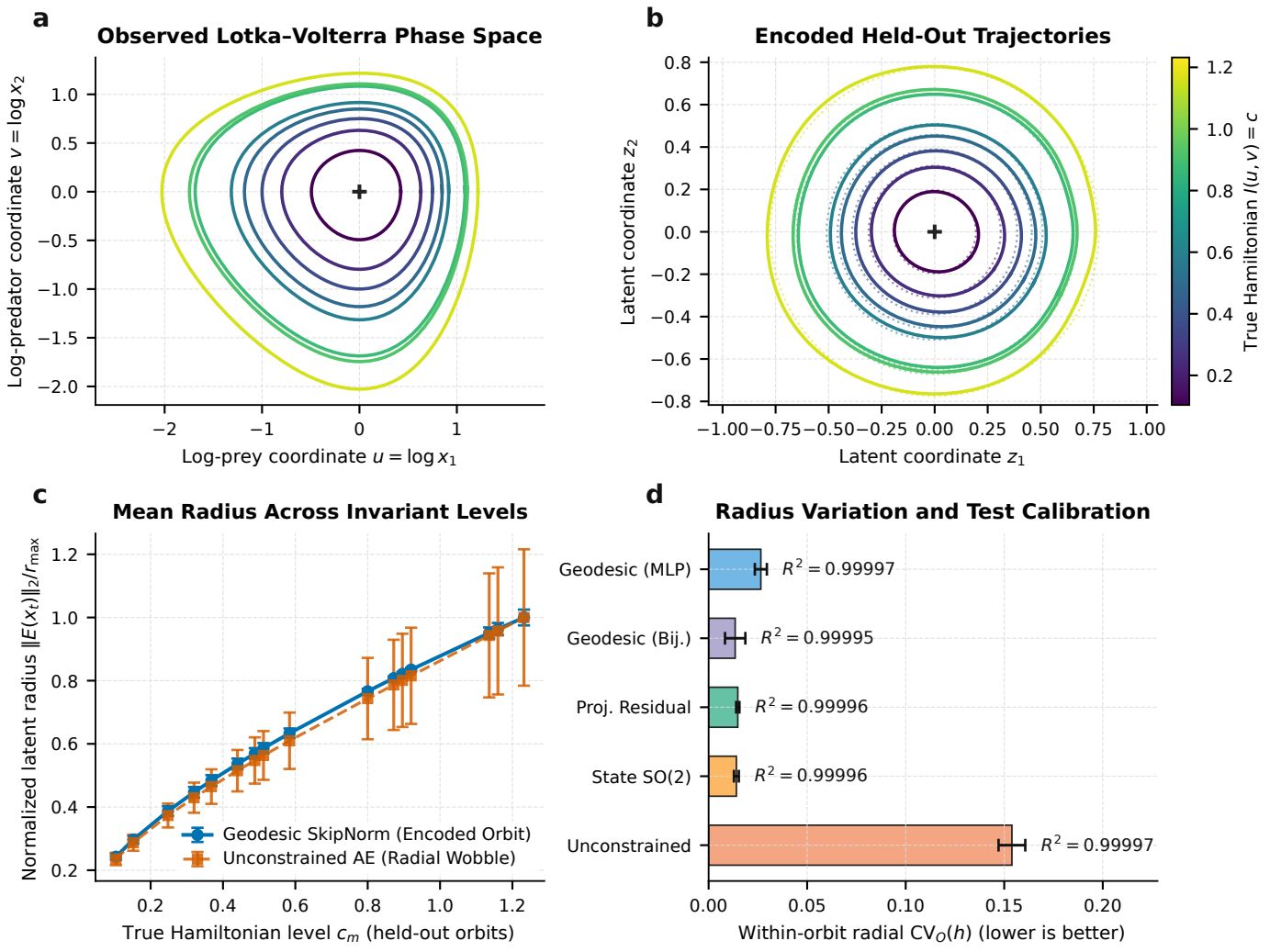

a
Observed Lotka–Volterra Phase Space
Log-predator coordinate $v = \log x_2$
Log-prey coordinate $u = \log x_1$
b
Encoded Held-Out Trajectories
Latent coordinate z_2
Latent coordinate z_1
True Hamiltonian $I(u, v) = c$
c
Mean Radius Across Invariant Levels
Normalized latent radius $\|E(x_t)\|_2 / r_{\max}$
Geodesic SkipNorm (Encoded Orbit)
Unconstrained AE (Radial Wobble)
True Hamiltonian level c_m (held-out orbits)
d
Radius Variation and Test Calibration
Geodesic (MLP)
Geodesic (Bij.)
Proj. Residual
State SO(2)
Unconstrained
$R^2 = 0.99997$
$R^2 = 0.99995$
$R^2 = 0.99996$
$R^2 = 0.99996$
$R^2 = 0.99997$
Within-orbit radial $\mathrm{CV}_O(h)$ (lower is better)